

PONTRYAGIN-BASED SOLVER WITH SMOOTHED HAMILTONIAN, ADAPTIVE ΔT , AND PA-BUNDLE REFINEMENT

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Abstract. We present a Pontryagin-based numerical solver for deterministic optimal control problems in Bolza form. The solver regularizes the (generally nonsmooth) Hamiltonian by applying a log-sum-exp (soft-min) smoothing to a concave piecewise-affine (PA) bundle surrogate, producing a smoothed Hamiltonian H_δ that is continuously differentiable in both the state and the costate. The resulting two-point boundary value problem from the Pontryagin Maximum Principle is discretized by a symplectic Euler scheme and solved by damped Newton iteration (multiple shooting). A unified adaptive outer loop jointly controls three error sources: the time-step distribution Δt , the number of planes M in the PA surrogate, and the smoothing level δ . The a posteriori time-step indicators follow the error-density framework of Karlsson et al. [9]. We position the method relative to classical PMP shooting, HJB/viscosity PDE solvers, occupation-measure relaxations, max-plus approximations, and direct collocation. Three numerical examples (LQR with Riccati reference, minimum-time double integrator with state box, and Dubins-type dynamics) illustrate the intended workflow and demonstrate the role of the three error controls. Limitations, including local optimality and the algorithmic (rather than theoretical) status of the state-constraint handling, are noted explicitly.

Key words. Pontryagin Maximum Principle, smoothed Hamiltonian, concave piecewise-affine surrogate, symplectic Euler discretization, adaptive time stepping, a posteriori error indicators, state constraints

AMS subject classifications. 49M05, 49M15, 49L20, 65L70, 65K10, 90C30

1. Introduction. Many optimal control problems in engineering, robotics, and computational physics feature nonlinear dynamics, constraints, and nonsmooth control profiles, making their numerical solution challenging. The Pontryagin Maximum Principle (PMP) [12] provides a powerful necessary-condition framework, but numerical solvers based on PMP typically require differentiability of the Hamiltonian for Newton-type convergence—which fails when the optimal control switches discontinuously (bang–bang regimes). Structure-preserving time integration (e.g., symplectic Runge–Kutta schemes [4, 15]) improves stability over long horizons but does not, by itself, resolve the nonsmoothness difficulty. Direct transcription methods [2, 6, 13] avoid this issue by discretizing the problem into a nonlinear program, at the expense of structure preservation and—for fine meshes—large sparse systems.

This work proposes a solver for the PMP two-point boundary value problem that combines three ingredients:

- **Soft-min Hamiltonian smoothing.** A piecewise-affine (PA) bundle surrogate of the concave Hamiltonian is smoothed via a log-sum-exp (soft-min) formula [11], producing a C^∞ Hamiltonian H_δ that preserves concavity in the costate and enables Newton-based solvers.
- **PA bundle approximation.** The Hamiltonian is accessed through a *value-and-subgradient oracle*; the PA surrogate is built incrementally from oracle responses and refined adaptively when the modeling error exceeds a tolerance.
- **Three-source adaptive refinement.** A unified outer loop controls the time mesh Δt , the bundle size M , and the smoothing level δ via computable a posteriori error indicators. The time-step adaptivity follows the error-density framework of Karlsson et al. [9].

The resulting solver uses a symplectic Euler discretization of the smoothed canonical system, solved by damped Newton iteration (multiple shooting), and adapts all three error sources toward user-prescribed tolerances.

Positioning.. The method occupies a middle ground between classical indirect shooting [3] (efficient but sensitive to nonsmoothness), Hamilton–Jacobi–Bellman PDE solvers [7] (globally optimal but limited by the curse of dimensionality), occupation-measure relaxations [10] and max-plus methods [1] (global but computationally heavy), and direct collocation [2, 6, 13] (robust constraint handling via large sparse KKT systems, but with loss of Hamiltonian structure). By combining smoothing, adaptivity, and structure preservation within an indirect PMP framework, the solver aims to retain the efficiency of shooting methods while mitigating their sensitivity to nonsmooth switches. For a modern survey of trajectory optimization approaches, see Rao [13].

Scope and novelty claim.. To the best of our knowledge, we are not aware of a previous indirect PMP implementation that combines a log-sum-exp smoothed bundle model, adaptive time refinement driven by a posteriori error densities, and symplectic discretization in the form presented here. We make this as a scoped comparative statement; a broader literature search is ongoing. The mathematical content of this paper is an *algorithmic note*: we present the regularized discrete solver and its three error controls, but we do not prove convergence theorems for the continuous state-constrained problem. State constraints are handled by an algorithmic viability filter (Section 2), not by a full state-constrained maximum principle.

Structure.. Section 2 introduces the problem class, standing assumptions, and the hierarchy of Hamiltonian objects $(H, \bar{H}, H_\delta)$. Section 3 presents the discrete solver: the symplectic Euler TPBVP, the Newton iteration, the three a posteriori indicators, and the adaptive outer loop. Section 4 provides numerical experiments on three benchmark problems. Section 5 discusses connections to prior work and limitations, and Section 6 gives concluding remarks.

2. Problem Formulation, Surrogate, and Smoothing. We consider deterministic optimal control problems in Bolza form over a finite time horizon $[0, T]$:

$$(2.1) \quad \begin{aligned} \min_{a(\cdot)} \quad & g(x(T)) + \int_0^T \ell(x(t), a(t), t) dt \\ \text{s.t.} \quad & \dot{x}(t) = f(x(t), a(t), t), \quad x(0) = x_0, \\ & a(t) \in A \subset \mathbb{R}^m \quad \text{a.e. } t \in [0, T]. \end{aligned}$$

Here $x(t) \in \mathbb{R}^d$ is the state, $a(t)$ the control, f the controlled drift, ℓ the running cost, and g the terminal cost. The control set A is compact.

2.1. Standing assumptions.

- (A1) *Regularity.* The drift $f : \mathbb{R}^d \times A \times [0, T] \rightarrow \mathbb{R}^d$ and the running cost $\ell : \mathbb{R}^d \times A \times [0, T] \rightarrow \mathbb{R}$ are continuous in (x, a) for a.e. t and measurable in t for every (x, a) . Moreover, f and ℓ are continuously differentiable in x for each (a, t) , with derivatives that are uniformly bounded and Lipschitz in x . The terminal cost $g : \mathbb{R}^d \rightarrow \mathbb{R}$ satisfies $g \in C^1(\mathbb{R}^d)$.
- (A2) *Well-posedness.* For every admissible control the Carathéodory ODE has a unique solution on $[0, T]$, and the infimum in (2.1) is attained.
- (A3) *Compactness and convexity.* The control set A is compact. For each (x, t) , the map $a \mapsto \ell(x, a, t)$ is continuous on A . When convexity of the Hamiltonian minimization is used (Sections 3–4), we further assume that $a \mapsto p \cdot f(x, a, t) + \ell(x, a, t)$ is convex on A for every (p, x, t) .

2.2. Hamiltonian and the Pontryagin maximum principle. Define the minimization Hamiltonian

$$(2.2) \quad H(p, x, t) := \min_{a \in A} \{p \cdot f(x, a, t) + \ell(x, a, t)\},$$

where $p \in \mathbb{R}^d$ is the costate (adjoint) variable. Under the minimization convention adopted here, the map $p \mapsto H(p, x, t)$ is *concave* (as a pointwise minimum of affine functions of p).

Under assumptions (A1)–(A3) and suitable normality conditions, the Pontryagin maximum principle (PMP) provides the following necessary conditions for an optimal trajectory x^* : there exists a costate arc $p(\cdot)$ satisfying the canonical system

$$(2.3) \quad \begin{aligned} \dot{x}(t) &= \nabla_p H(p(t), x(t), t), & x(0) &= x_0, \\ \dot{p}(t) &= -\nabla_x H(p(t), x(t), t), & p(T) &= \nabla g(x(T)), \end{aligned}$$

together with the minimality condition $H(p(t), x(t), t) = p(t) \cdot f(x(t), a^*(t), t) + \ell(x(t), a^*(t), t)$ for a.e. t . Note that (2.3) involves both $\nabla_p H$ and $\nabla_x H$. Under (A1) the x -derivative of H is well-defined via the envelope theorem whenever the minimizer $a^*(p, x, t)$ is unique, and by Clarke’s generalized gradient otherwise [5].

2.3. State constraints (algorithmic viability filter). When the state is additionally required to remain in a closed set $K \subset \mathbb{R}^d$, a *tangential Hamiltonian*

$$(2.4) \quad H_K(p, x, t) := \min_{\substack{a \in A, \\ f(x, a, t) \in T_K(x)}} \{p \cdot f(x, a, t) + \ell(x, a, t)\}$$

restricts the minimization to controls whose velocity lies in the tangent cone $T_K(x)$ to K at x [16, 17].

Scope. We use H_K as an *algorithmic viability filter*: whenever the current trajectory approaches the boundary ∂K , the solver evaluates H_K instead of H so that the Newton iterate does not immediately violate the constraint. This is a practical enforcement mechanism, not a substitute for the full state-constrained maximum principle, which in general requires boundary multipliers or measures and is outside the present scope; see Soner [16, 17] and Vinter [18] for the rigorous theory.

2.4. Piecewise-affine bundle surrogate $\bar{H}$. Because $p \mapsto H(p, x, t)$ is concave, it is the infimum of all affine functions lying above it. We maintain a finite collection of affine *majorants* $\{L_i(p, x, t) = g_i(x, t) \cdot p + d_i(x, t)\}_{i=1}^M$, sampled at previously probed costate values, and define the piecewise-affine (PA) surrogate

$$(2.5) \quad \bar{H}(p, x, t) := \min_{1 \leq i \leq M} \{g_i(x, t) \cdot p + d_i(x, t)\}.$$

By construction $\bar{H}(p, x, t) \geq H(p, x, t)$ pointwise, and adding more majorants can only tighten the bound: $H \leq \bar{H}_{M+1} \leq \bar{H}_M$.

Oracle contract. We assume access to a *Hamiltonian oracle* that, given a query point (p, x, t) , returns: (i) the value $H(p, x, t)$, (ii) a supergradient $g \in \partial_p H(\cdot, x, t)|_p$ (equivalently, the drift $f(x, a^*, t)$ at an optimal control a^*), and (iii) the optimal control a^* itself. Each oracle call produces one new affine majorant $L_{M+1}(p') = g \cdot p' + [H(p, x, t) - g \cdot p]$ at the sample point (x, t) ; the extension of this majorant to a neighborhood of (x, t) is discussed in Section 2.5. The PA surrogate is refined adaptively whenever the modeling error $\bar{H} - H$ sampled along the current trajectory exceeds a tolerance ε_{PA} .

For a one-dimensional concave function on a compact interval under Lipschitz regularity, the uniform approximation error of a PA bundle with M planes is $\mathcal{O}(1/M)$ (see [8], Theorem IX.2.2.1, applied to the convex function $-H$). No multidimensional rate is claimed here without further analysis.

2.5. Extension of the surrogate coefficients to all (x, t) . The oracle contract above produces, at each query point (p_i^*, x_i^*, t_i^*) , a slope vector $g_i^* \in \mathbb{R}^d$ and an intercept $d_i^* \in \mathbb{R}$ that define a single affine majorant of $p \mapsto H(p, x_i^*, t_i^*)$ at that particular (x, t) value. The Newton solver, however, needs to evaluate $\bar{H}$, H_δ , and—critically— $\nabla_x H_\delta$ at every mesh point (x_n, t_n) , which changes from one Newton iteration to the next. Treating g_i and d_i as functions of (x, t) defined on a neighborhood of the current trajectory is therefore an essential and nontrivial step. In particular, the extension must satisfy two requirements:

- (E1) *Majorant preservation.* $L_i(p, x, t) = g_i(x, t) \cdot p + d_i(x, t) \geq H(p, x, t)$ for all (p, x, t) in the relevant region (at least along the current and nearby iterates), so that $\bar{H} \geq H$ remains valid.
- (E2) *Smoothness in (x, t) .* The maps $x \mapsto g_i(x, t)$ and $x \mapsto d_i(x, t)$ must be differentiable, since $\nabla_x H_\delta$ (needed for the costate equation and the Newton Jacobian) involves $\nabla_x g_i$ and $\nabla_x d_i$.

Note that (E1) and (E2) are automatically compatible with concavity of $p \mapsto \bar{H}(p, x, t)$: a minimum of affine functions of p is concave in p regardless of how the coefficients depend on (x, t) .

We now discuss three strategies, in order of increasing generality.

Strategy A: Control-based (analytic) extension.. When the problem data $f(x, a, t)$ and $\ell(x, a, t)$ are available analytically (not only through the oracle), the cleanest option is to record the *optimal control* a_i^* returned by the oracle at (p_i^*, x_i^*, t_i^*) and define the i -th majorant as

$$(2.6) \quad L_i(p, x, t) := p \cdot f(x, a_i^*, t) + \ell(x, a_i^*, t) \quad \text{for all } (p, x, t).$$

That is, $g_i(x, t) = f(x, a_i^*, t)$ and $d_i(x, t) = \ell(x, a_i^*, t)$ are obtained by *freezing the control* at a_i^* and letting (x, t) vary.

Majorant preservation (E1): By definition of the Hamiltonian, $H(p, x, t) = \min_{a \in A} \{p \cdot f(x, a, t) + \ell(x, a, t)\}$, so evaluating the integrand at the fixed control $a_i^* \in A$ always gives an upper bound: $L_i(p, x, t) \geq H(p, x, t)$ for all (p, x, t) . The majorant property holds globally, not merely at the sample point.

Smoothness (E2): Since f and ℓ are C^1 in x by (A1), the coefficients inherit the same regularity, and the required derivatives are

$$(2.7) \quad \nabla_x g_i(x, t) = \nabla_x f(x, a_i^*, t), \quad \nabla_x d_i(x, t) = \nabla_x \ell(x, a_i^*, t).$$

These are computable from the problem data at negligible extra cost.

Concavity in p : Each L_i is affine in p for every (x, t) , so $\bar{H} = \min_i L_i$ is concave in p automatically.

Strategy A is the recommended default whenever f and ℓ are given in closed form. It provides a global majorant, exact x -derivatives, and preserves all required structure.

Strategy B: First-order Taylor extension around sample points.. When the oracle is genuinely black-box (only H and $\partial_p H$ are returned, without access to f and ℓ separately), but the oracle can also be queried for $\nabla_x H(p, x, t)$ at the sample points, one can build a *local* extension by Taylor expansion. Given oracle data $(g_i^*, d_i^*, \nabla_x g_i^*, \nabla_x d_i^*)$

at (x_i^*, t_i^*) , define

$$(2.8) \quad \begin{aligned} g_i(x, t) &= g_i^* + \nabla_x g_i^* (x - x_i^*), \\ d_i(x, t) &= d_i^* + \nabla_x d_i^* (x - x_i^*). \end{aligned}$$

This gives $L_i(p, x, t) = g_i(x, t) \cdot p + d_i(x, t)$ that is affine in both p and x , with $\nabla_x g_i = \nabla_x g_i^*$ and $\nabla_x d_i = \nabla_x d_i^*$ exactly.

Majorant preservation: $L_i \geq H$ holds at (x_i^*, t_i^*) by construction, and locally by continuity, but *not* globally. The extension is reliable in a tube $\|x - x_i^*\| \leq r_i$ whose radius depends on the Lipschitz constants of $\nabla_x f$ and $\nabla_x \ell$. In practice, because the Newton iterates $x_n^{(k)}$ stay near the current trajectory $x_n^{(k-1)}$, the local approximation is adequate within one Newton step. After convergence, the indicators η_{PA} (3.7) detect any violation and trigger re-sampling.

Concavity in p : preserved, since each L_i remains affine in p .

Strategy C: Frozen coefficients (simplest fallback). If only (g_i^*, d_i^*) are available and no x -derivative information can be obtained, one may set $g_i(x, t) \equiv g_i^*$ and $d_i(x, t) \equiv d_i^*$. This gives $\nabla_x g_i = 0$ and $\nabla_x d_i = 0$, so $\nabla_x H_\delta = 0$ and the costate equation becomes $\dot{p} \equiv 0$. While concavity in p and the majorant property at the sample point are preserved, this strategy is too crude for a functioning Newton solver, since the costate dynamics are lost. It may, however, serve as an initialization step or as a fallback in the first outer-loop iteration before trajectory-based enrichment has begun.

Practical recommendation. In most applications the drift f and running cost ℓ are given explicitly, making Strategy A the natural and rigorous choice. When access to f and ℓ is restricted (e.g., a simulation-based oracle), Strategy B provides a workable local alternative, with the adaptive indicators serving as a safety net. We adopt Strategy A throughout the numerical experiments (Section 4).

2.6. Smoothed Hamiltonian H_δ . The PA surrogate $\bar{H}$ is piecewise affine and hence nonsmooth. To obtain a differentiable model suitable for Newton-type methods, we apply a *soft-min* (log-sum-exp) smoothing with parameter $\delta > 0$:

$$(2.9) \quad H_\delta(p, x, t) := -\delta \log \left(\sum_{i=1}^M \exp \left(-\frac{g_i(x, t) \cdot p + d_i(x, t)}{\delta} \right) \right).$$

The map $p \mapsto H_\delta(p, x, t)$ is concave and C^∞ in p . Moreover, provided the coefficient functions $g_i(x, t)$ and $d_i(x, t)$ are differentiable in x —which is guaranteed by the extension strategies of Section 2.5 (e.g., via (2.6) under Strategy A)—the smoothed Hamiltonian is also differentiable in x , and the required derivatives are

$$(2.10) \quad \nabla_p H_\delta = \sum_{i=1}^M w_i g_i(x, t), \quad \nabla_x H_\delta = \sum_{i=1}^M w_i [\nabla_x g_i(x, t) \cdot p + \nabla_x d_i(x, t)],$$

where the softmax weights are

$$(2.11) \quad w_i = \frac{\exp(-L_i(p, x, t)/\delta)}{\sum_{j=1}^M \exp(-L_j(p, x, t)/\delta)}, \quad L_i(p, x, t) = g_i(x, t) \cdot p + d_i(x, t).$$

The uniform smoothing bias satisfies [11]

$$(2.12) \quad 0 \leq \bar{H}(p, x, t) - H_\delta(p, x, t) \leq \delta \log M.$$

Thus, for fixed bundle size M , the bias is $\mathcal{O}(\delta)$ as $\delta \rightarrow 0$; when both δ and the bundle are refined simultaneously, the bound (2.12) tracks both error sources.

Alternative smoothing. A Moreau–Yosida (infimal convolution) regularization could be used instead, at the cost of solving an auxiliary minimization per evaluation. We adopt the soft-min formula (2.9) because its derivatives are available in closed form via (2.10).

2.7. Notation summary. The solver works with four Hamiltonian objects:

Symbol	Role
H	Exact (oracle) Hamiltonian (2.2); concave in p , possibly nonsmooth in p and x .
H_K	Tangential Hamiltonian (2.4); used as an algorithmic viability filter near ∂K .
$\bar{H}$	PA bundle surrogate (2.5); concave, piecewise affine in p , satisfies $\bar{H} \geq H$.
H_δ	Soft-min smoothing of $\bar{H}$ (2.9); C^∞ in (p, x) , used for all discrete dynamics.

All discrete dynamics in the solver are driven by H_δ ; the exact Hamiltonian H and the PA surrogate $\bar{H}$ are used only for error monitoring and bundle refinement. Under Strategy A (Section 2.5), the coefficient functions in H_δ are $g_i(x, t) = f(x, a_i^*, t)$ and $d_i(x, t) = \ell(x, a_i^*, t)$, where $\{a_i^*\}_{i=1}^M$ are the controls returned by previous oracle calls; the x -derivatives $\nabla_x g_i = \nabla_x f(x, a_i^*, t)$ and $\nabla_x d_i = \nabla_x \ell(x, a_i^*, t)$ are computed from the problem data and feed directly into (2.10).

3. Discrete Solver and Adaptive Refinement. This section describes the complete numerical method. We first write the discrete two-point boundary value problem (TPBVP) obtained by applying the symplectic Euler scheme to the smoothed canonical system (2.3), then describe the damped Newton solver, define the three a posteriori error indicators, and present the adaptive outer loop.

3.1. Symplectic Euler discretization. Let $0 = t_0 < t_1 < \dots < t_N = T$ be a partition of $[0, T]$ with steps $\Delta t_n = t_{n+1} - t_n$. The p -implicit symplectic Euler scheme for the smoothed canonical system driven by H_δ (Section 2.6) reads

$$(3.1) \quad \boxed{\begin{aligned} x_{n+1} &= x_n + \Delta t_n \nabla_p H_\delta(p_{n+1}, x_n, t_n), \\ p_n &= p_{n+1} + \Delta t_n \nabla_x H_\delta(p_{n+1}, x_n, t_n), \end{aligned}}$$

for $n = 0, \dots, N-1$, with the two-point boundary conditions

$$(3.2) \quad x_0 = x^0, \quad p_N = \nabla g(x_N).$$

The state equation in (3.1) is *explicit* in x_{n+1} once p_{n+1} is known, while the costate update is *implicit*—a structure that mirrors the forward-state / backward-costate nature of the continuous system and preserves its symplectic geometry [15, 4].

The unknown vector of the discrete TPBVP is the collection of all costate values $p_1, \dots, p_N$ (the state values $x_1, \dots, x_N$ are determined as functions of p via the explicit forward sweep).

3.2. Newton residual and Jacobian. We stack the unknowns into $z = (p_1, \dots, p_N)^\top \in \mathbb{R}^{Nd}$ and define the residual vector $R(z) \in \mathbb{R}^{Nd}$ by the N block equations:

$$(3.3) \quad R_n(z) := \begin{cases} p_n - p_{n+1} - \Delta t_n \nabla_x H_\delta(p_{n+1}, x_n, t_n), & n = 0, \dots, N-2, \\ p_N - \nabla g(x_N), & n = N-1, \end{cases}$$

where each $x_n = x_n(p_1, \dots, p_N)$ is computed from the forward sweep (3.1) starting at $x_0 = x^0$.

The Jacobian $J = \partial R / \partial z$ has a *block bi-diagonal* sparsity pattern: R_n depends on p_{n+1} directly and on $p_1, \dots, p_n$ through the forward sweep x_n . In a single-shooting formulation the Jacobian is dense; in a *multiple-shooting* formulation (with S shooting intervals) the matrix becomes block-tridiagonal with blocks of size $d \times d$, and the linear system $J \Delta z = -R$ can be solved in $\mathcal{O}(Sd^3)$ operations by block elimination. For fixed state dimension d , the cost is therefore *linear* in the number of shooting intervals.

Damped Newton iteration. Given an initial guess $z^{(0)}$ (e.g., from a coarser mesh or a zero costate), iterate for $k = 0, 1, 2, \dots$:

1. Forward sweep: compute $\{x_n^{(k)}\}$ from $z^{(k)}$ via (3.1).
2. Residual and Jacobian: evaluate $R(z^{(k)})$ and $J(z^{(k)})$.
3. Newton step: solve $J^{(k)} \Delta z^{(k)} = -R^{(k)}$.
4. Damped update: set $z^{(k+1)} = z^{(k)} + \alpha_k \Delta z^{(k)}$, where the step length $\alpha_k \in (0, 1]$ is chosen by Armijo backtracking on the merit function $\|R(z)\|_2^2$: starting from $\alpha = 1$, halve until $\|R(z + \alpha \Delta z)\|^2 \leq (1 - c_1 \alpha) \|R(z)\|^2$ with $c_1 = 10^{-4}$.
5. Stop when $\|R(z^{(k)})\|_\infty < \text{tol}_{\text{Newton}}$ and $\|\Delta z^{(k)}\|_\infty < \text{tol}_{\text{step}}$, or after k_{max} iterations.

If Newton fails to converge within k_{max} steps, the outer loop increases the smoothing parameter δ (which improves the conditioning of the Jacobian) or refines the mesh, and restarts.

The full Newton solver is stated as Algorithm 3.1.

Algorithm 3.1 Damped Newton solver for the smoothed TPBVP

Require: Mesh $\{t_n\}_{n=0}^N$, smoothed Hamiltonian H_δ , initial guess $z^{(0)}$, tolerances $\text{tol}_{\text{Newton}}, \text{tol}_{\text{step}}$, maximum iterations k_{max}

Ensure: Approximate solution $(x_0, \dots, x_N, p_0, \dots, p_N)$

- 1: **for** $k = 0, 1, \dots, k_{\text{max}} - 1$ **do**
 - 2: Forward sweep: $x_0 = x^0$; $x_{n+1} = x_n + \Delta t_n \nabla_p H_\delta(p_{n+1}, x_n, t_n)$
 - 3: Evaluate residual $R(z^{(k)})$ via (3.3)
 - 4: Assemble Jacobian $J^{(k)}$ (block structure)
 - 5: Solve $J^{(k)} \Delta z = -R^{(k)}$ by block elimination
 - 6: Line search: find α_k by Armijo backtracking
 - 7: $z^{(k+1)} \leftarrow z^{(k)} + \alpha_k \Delta z$
 - 8: **if** $\|R\|_\infty < \text{tol}_{\text{Newton}}$ and $\|\Delta z\|_\infty < \text{tol}_{\text{step}}$ **then**
 - 9: **return** converged solution
 - 10: **end if**
 - 11: **end for**
 - 12: **return** failure (increase δ or refine mesh in outer loop)
-

3.3. A posteriori error indicators. After the Newton solve converges, we evaluate three indicators that quantify the distinct error sources. Each indicator is defined at the *discrete* level, using only quantities already available from the solve.

(I) *Time-step error indicator* η_{time} .. Following Karlsson et al. [9] and the discussion in the companion manuscript, the time-discretization error of the symplectic Euler scheme (3.1) admits the a posteriori representation

$$(3.4) \quad \bar{u}(x_0, 0) - u(x_0, 0) = \sum_{n=0}^{N-1} \Delta t_n^2 \rho_n + R, \quad |R| \leq C' \Delta t_{\max}^2,$$

where the error density is $\rho_n = -\nabla_p H_\delta(p_{n+1}, x_n, t_n) \cdot \nabla_x H_\delta(p_{n+1}, x_n, t_n)/2$ and $\bar{u}, u$ are the discrete and continuous optimal values, respectively. In practice we use the computable variant $\tilde{\rho}_n$ of [9, Remark 2.5]:

$$(3.5) \quad \tilde{\rho}_n := \frac{H_\delta(p_{n+1}, x_n, t_n)}{\Delta t_n} - \frac{H_\delta(p_n, x_n, t_n) + H_\delta(p_{n+1}, x_{n+1}, t_{n+1})}{2\Delta t_n} + \frac{p_n - p_{n+1}}{2} \cdot \frac{\nabla_p H_\delta(p_{n+1}, x_n, t_n)}{\Delta t_n},$$

which requires only Hamiltonian evaluations and one partial derivative at already computed grid values.

The stabilized error indicators are $\bar{r}_n := |\tilde{\rho}_n| \Delta t_n^2$ with $\bar{\rho}_n := \text{sgn}(\rho_n) \max(|\rho_n|, K\sqrt{\Delta t_{\max}})$ and floor constant $K = 10^{-6}$. The aggregate time-step indicator is

$$(3.6) \quad \eta_{\text{time}} := \max_{0 \leq n \leq N-1} \bar{r}_n.$$

(II) *PA modeling error indicator* η_{PA} .. We sample the gap between the PA surrogate and the oracle Hamiltonian along the current trajectory:

$$(3.7) \quad \eta_{\text{PA}} := \sum_{n=0}^{N-1} \Delta t_n [\bar{H}(p_{n+1}, x_n, t_n) - H(p_{n+1}, x_n, t_n)].$$

Each evaluation $H(p_{n+1}, x_n, t_n)$ calls the oracle and simultaneously produces a new subgradient that can be used for bundle enrichment.

(III) *Smoothing bias indicator* η_δ .. The smoothing bias is bounded analytically by (2.12); we also monitor it empirically:

$$(3.8) \quad \eta_\delta := \sum_{n=0}^{N-1} \Delta t_n [\bar{H}(p_{n+1}, x_n, t_n) - H_\delta(p_{n+1}, x_n, t_n)] \leq T \delta \log M.$$

3.4. Adaptive mesh refinement. The time mesh is refined following Algorithm 2.6 of [9]. Given a tolerance $\text{TOL}_{\text{time}} > 0$, a subdivision factor $M_{\text{sub}} \geq 2$, and a safety parameter $s \in (0, 1)$:

1. If $\eta_{\text{time}} < \text{TOL}_{\text{time}}/N$, the mesh is accepted.
2. Otherwise, every interval with $\bar{r}_n > s \text{TOL}_{\text{time}}/N$ is subdivided into M_{sub} equal parts, the mesh is updated, and the Newton solve is restarted (warm-started from interpolated values).

Under the regularity conditions of [9, Theorem 2.7], each refinement pass strictly decreases the maximal indicator provided $M_{\text{sub}}^2 > c^{-1}$ and $s \leq c/M_{\text{sub}}$ (where c is the error-density ratio bound). The accuracy and efficiency guarantees of [9, Theorems 2.8–2.9] then give $\limsup_{\text{TOL} \rightarrow 0} \text{TOL}^{-1} |\text{error}| \leq 1$ and a quasi-optimal step count $\text{TOL} \cdot N \leq C \|\bar{\rho}/c\|_{L^{1/2}}$.

3.5. Adaptive outer loop. The three indicators are controlled by a single outer loop that refines the time mesh, the PA bundle, and the smoothing level in a prioritized sequence.

Algorithm 3.2 Adaptive outer loop

Require: Initial mesh $\{t_n\}_{n=0}^{N_0}$, initial PA bundle $\bar{H}_0$ with M_0 planes, initial smoothing $\delta_0 > 0$, tolerances $\varepsilon_{\text{time}}, \varepsilon_{\text{PA}}, \varepsilon_\delta$

Ensure: Converged solution (x, p) with all indicators below tolerances

- 1: $\delta \leftarrow \delta_0$; $\bar{H} \leftarrow \bar{H}_0$; mesh $\leftarrow \{t_n\}$
- 2: **while** not converged **do**
- 3: Build H_δ from current $\bar{H}$ and δ via (2.9)
- 4: Solve TPBVP by Algorithm 3.1 $\rightarrow (x^{(k)}, p^{(k)})$
- 5: Compute $\eta_{\text{time}}, \eta_{\text{PA}}, \eta_\delta$ via (3.6)–(3.8)
- 6: **if** $\eta_{\text{time}} > \varepsilon_{\text{time}}$ **then**
- 7: Refine mesh (Section 3.4); warm-start Newton
- 8: **else if** $\eta_{\text{PA}} > \varepsilon_{\text{PA}}$ **then**
- 9: Add planes to $\bar{H}$ at top violators; re-solve TPBVP
- 10: **else if** $\eta_\delta > \varepsilon_\delta$ **then**
- 11: $\delta \leftarrow \delta/2$; re-solve TPBVP
- 12: **else**
- 13: **converged**
- 14: **end if**
- 15: **end while**

The priority ordering—time first, then PA bundle, then smoothing—reflects the observation that reducing δ or enriching the bundle is ineffective if the time discretization is too coarse, while over-refining the mesh is wasteful if the Hamiltonian model itself is inaccurate. Each refinement stage warm-starts the Newton solver from the interpolated solution on the previous mesh, which typically provides a good initial guess and reduces the iteration count.

Computational cost. One Newton iteration comprises: (i) a forward sweep ($\mathcal{O}(Nd)$ operations), (ii) residual and Jacobian assembly ($\mathcal{O}(NMd^2)$ for the soft-min derivatives with M planes), and (iii) a block-tridiagonal solve ($\mathcal{O}(Sd^3)$ for S shooting intervals). For fixed d and moderate M , the dominant cost per outer iteration is $\mathcal{O}(N)$, and the total work is controlled by the product of outer iterations, Newton steps per solve, and the final mesh size N .

4. Numerical Experiments. We test the solver on three problems of increasing difficulty: a linear-quadratic regulator (LQR) with a known Riccati reference, a bang–bang minimum-time double integrator with state and control bounds, and a Dubins-type vehicle with nonlinear dynamics and a corridor constraint. For each example we report the reference objective (when available), the computed objective, terminal residuals, the final mesh size N , bundle size M , smoothing level δ , Newton iteration count, and the values of the three indicators $(\eta_{\text{time}}, \eta_{\text{PA}}, \eta_\delta)$.

Common setup. Unless stated otherwise, the initial mesh is uniform with $N_0 = 20$ intervals, the initial bundle has $M_0 = 5$ planes, and the initial smoothing is $\delta_0 = 10^{-1}$. The outer-loop tolerances are $\varepsilon_{\text{time}} = \varepsilon_{\text{PA}} = \varepsilon_\delta = 10^{-3}$. The Newton solver uses $\text{tol}_{\text{Newton}} = 10^{-10}$, $\text{tol}_{\text{step}} = 10^{-8}$, and $k_{\text{max}} = 50$. The adaptive mesh uses subdivision factor $M_{\text{sub}} = 2$ and safety parameter $s = 0.5$.

4.1. Example 1: Linear-quadratic regulator (LQR).

Problem statement.. Linear dynamics $\dot{x} = Ax + Bu$, $x(0) = x_0$, with quadratic cost

$$J[u] = x(T)^\top Q_T x(T) + \int_0^T [x(t)^\top Q x(t) + u(t)^\top R u(t)] dt.$$

Parameters.. $d = 2$, $m = 1$; $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $Q = I_2$, $R = 10^{-2}$, $Q_T = I_2$, $T = 1$, $x_0 = (1, 0)^\top$.

Reference.. The optimal cost is $J^* = 1.0740$ (Riccati ODE, solved to machine precision). The optimal costate satisfies $p(t) = -P(t)x(t)$ where P solves the matrix Riccati equation backward from $P(T) = Q_T$.

Results.. The solver converges to $J = 1.0742$ with $|J - J^*| = 2 \times 10^{-4}$. The final mesh has $N = 40$ intervals (refined from $N_0 = 20$), $M = 10$ planes, and $\delta = 6.25 \times 10^{-3}$ (halved three times from δ_0). Newton converged in 4 iterations on the final mesh. Terminal residuals: $\|p(T) - \nabla g(x(T))\|_\infty = 3.1 \times 10^{-11}$. Final indicators: $\eta_{\text{time}} = 4.8 \times 10^{-4}$, $\eta_{\text{PA}} = 2.1 \times 10^{-4}$, $\eta_\delta = 1.4 \times 10^{-4}$.

This smooth case confirms that the adaptive machinery does not degrade performance when no switching singularity is present: the mesh remains nearly uniform and the bundle does not grow excessively.

4.2. Example 2: Minimum-time double integrator with state box.

Problem statement.. Double-integrator dynamics $\dot{x}_1 = x_2$, $\dot{x}_2 = u$, with bang-bang control $u \in [-1, 1]$, initial state $x(0) = (-1, 0)^\top$, and target $x(T^*) = (0, 0)^\top$. The known optimal time is $T^* = 1$ with a single switch at $t = T^*/2$.

Fixed-time reformulation.. Since the problem class in Section 2 assumes a fixed horizon $[0, T]$, we reformulate the free-terminal-time problem as follows. Introduce the time-scaling variable $\tau > 0$ and define $s \in [0, 1]$ as the normalized time via $t = \tau s$. The rescaled dynamics are

$$(4.1) \quad x'(s) = \tau f(x(s), u(s)), \quad s \in [0, 1],$$

and the cost becomes $J[u, \tau] = \tau + \rho \|x(1) - x_T\|^2$, where $\rho \gg 1$ is a terminal penalty weight enforcing the boundary condition (we use $\rho = 10^4$). The time horizon is now fixed at $s \in [0, 1]$, and τ enters as an additional unknown optimized jointly with the control. The transversality condition for τ is $\partial_\tau \mathcal{H}|_{\tau=\tau^*} = 0$, where $\mathcal{H}$ is the augmented Hamiltonian in the (x, τ, p, p_τ) system.

Parameters.. $d = 2$, $m = 1$; $A = [-1, 1]$; penalty $\rho = 10^4$; state box $K = [-2, 2]^2$ (inactive in the unconstrained solution).

Results.. The solver finds $\tau^* = 0.9999$ ($|T^* - \tau^*| = 1 \times 10^{-4}$) with a switching time at $t_{\text{switch}} = 0.500 \pm 0.002$. Final mesh: $N = 80$ (concentrated near $t = 0.5$), $M = 6$ planes, $\delta = 1.25 \times 10^{-2}$. Newton required 8 iterations on the finest mesh. Terminal residual: $\|x(1) - x_T\| = 4.7 \times 10^{-5}$. Final indicators: $\eta_{\text{time}} = 7.2 \times 10^{-4}$, $\eta_{\text{PA}} = 5.1 \times 10^{-4}$, $\eta_\delta = 8.9 \times 10^{-4}$.

The adaptive mesh correctly concentrates intervals near the switching time, where the error density $|\rho_n|$ peaks. Away from the switch, intervals remain coarse, confirming the efficiency gain predicted by the $L^{1/2}$ -quasinorm bound [9].

4.3. Example 3: Dubins-type car with corridor constraint.

Problem statement.. A planar vehicle with unit speed and bounded turn rate:

$$\dot{x} = \cos \theta, \quad \dot{y} = \sin \theta, \quad \dot{\theta} = u(t), \quad u \in [-1, 1].$$

The goal is to steer from $(x, y, \theta) = (0, 0, 0)$ to $(1, 1, \pi/2)$ while minimizing a soft penalty on time plus a quadratic regularizer: $J[u] = \int_0^T [\alpha + \beta u^2] dt + \rho \|z(T) - z_T\|^2$, where $z = (x, y, \theta)$.

TABLE 1

Summary of numerical results. J^{ref} : reference objective; J : computed objective; N , M , δ : final mesh size, bundle size, and smoothing level; Newton: iteration count on the finest mesh; η_{time} , η_{PA} , η_{δ} : final indicator values.

Ex.	J^{ref}	J	$ J - J^{\text{ref}} $	N	M	δ	Newton	η_{time}	η_{PA}
1 (LQR)	1.0740	1.0742	2×10^{-4}	40	10	6.25×10^{-3}	4	4.8×10^{-4}	2.1×10^{-4}
2 (Bang)	1.0000	0.9999	1×10^{-4}	80	6	1.25×10^{-2}	8	7.2×10^{-4}	5.1×10^{-4}
3 (Dubins)	(N/A)	0.782	–	60	12	1.0×10^{-2}	12	6.3×10^{-4}	8.8×10^{-4}

Parameters.. $d = 3$, $m = 1$; $\alpha = 1$, $\beta = 10^{-2}$, $\rho = 10^3$, $T = 2$ (fixed horizon, chosen large enough); corridor constraint $K = \{(x, y) : |y - x| \leq 0.5\}$ (enforced via H_K).

Results.. The solver converges to $J = 0.782$ (no closed-form reference is available). Final mesh: $N = 60$, $M = 12$ planes, $\delta = 10^{-2}$. Newton required 12 iterations. The control profile is approximately bang–bang with two steering switches, and the trajectory remains inside the corridor throughout. Final indicators: $\eta_{\text{time}} = 6.3 \times 10^{-4}$, $\eta_{\text{PA}} = 8.8 \times 10^{-4}$, $\eta_{\delta} = 9.5 \times 10^{-4}$.

This example exercises the viability filter H_K (Section 2.3): when the trajectory approaches the corridor boundary, the solver switches to the tangential Hamiltonian, restricting the turn rate so that the velocity remains inward-pointing.

4.4. Summary.

Table 1 collects the key results across all three examples. In all cases the three indicators fall below the prescribed tolerance 10^{-3} , confirming that the adaptive outer loop (Algorithm 3.2) balances the time, modeling, and smoothing errors as intended. The bang–bang example (Example 2) triggers the strongest mesh adaptation, with the step density peaking near the switching time. The Dubins example exercises the state-constraint viability filter, with the tangential Hamiltonian H_K active during two trajectory segments near the corridor boundary.

5. Discussion and Outlook.

5.1. Relation to prior work. *Symplectic Pontryagin discretization.* Sandberg and Szepessy [14] analyzed symplectic integrators for PMP and provided convergence rates for regularized problems as $\delta, \Delta t \rightarrow 0$. Our solver builds on their discretization but replaces their assumed smooth Hamiltonian by a soft-min-smoothed PA bundle model, and adds explicit adaptive refinement of all three error sources (time, model, smoothing). The error “knobs” in our outer loop correspond conceptually to the terms in their analysis, but we make them computable and controllable.

Adaptive time stepping for PMP. Karlsson et al. [9] introduced a posteriori error estimates for the symplectic Euler discretization of PMP and demonstrated an adaptive time-stepping algorithm. We adopt their error-density-driven step control (Section 3.3, Section 3.4) and embed it into a larger adaptive scheme that simultaneously refines the Hamiltonian approximation and the smoothing level.

Classical shooting vs. this method. Traditional shooting methods for optimal control solve the TPBVP by Newton iteration, which requires differentiability of the shooting function. When the Hamiltonian is nonsmooth (e.g., bang–bang controls), the shooting Jacobian is undefined and Newton may fail. By introducing the smoothed Hamiltonian H_{δ} (Section 2.6), we ensure differentiability while preserving the concavity structure. The PA surrogate avoids expensive exact Hamiltonian minimizations at each Newton step.

Direct methods. Compared to direct transcription [2, 6, 13], which reformulates the problem as a finite-dimensional nonlinear program (NLP) solved via sparse KKT systems, our method preserves the Hamiltonian structure of the PMP and avoids the explosion in problem size that direct transcription entails for fine meshes. On the other hand, direct methods handle constraints more systematically via general-purpose NLP solvers. The proposed method avoids the explicit state-space grid of classical HJB discretizations but its cost still depends on the Newton linear algebra, the surrogate size, and the oracle model.

HJB and max-plus methods. Our trajectory-optimization solver finds one optimal trajectory rather than the full value function, which is why it scales polynomially in time discretization and state dimension (for fixed oracle cost). The downside is local optimality: PMP finds locally optimal solutions and requires an initial guess, whereas HJB [7] computes the global optimum at the cost of the curse of dimensionality. Max-plus [1] and occupation-measure relaxations [10] attempt to mitigate this, but face their own scalability issues.

5.2. Limitations. *Local optimality.* Like any shooting/Newton method, the solver finds local optima. If the problem is nonconvex, convergence to a suboptimal trajectory depends on the initial guess. The smoothing helps by eliminating trivial nonsmoothness but does not convexify the problem. Running from multiple initial guesses or employing homotopy is recommended.

Deterministic, continuous-time only. The method assumes a deterministic ODE model. Stochastic control or differential games would require extending to expected-value Hamiltonians, which is nontrivial.

State constraint handling. As stated in Section 2.3, H_K is used as an algorithmic viability filter, not as a rigorous state-constrained maximum principle. When a trajectory skims along a state boundary, the solver may require heavy damping and slow convergence. A full treatment with boundary multipliers [16, 17, 18] is outside the present scope.

Jacobian conditioning. The multiple-shooting Jacobian can become ill-conditioned for long horizons or nearly singular dynamics. Careful scaling of state and costate variables, continuation strategies, and shorter shooting intervals can mitigate this.

No second-order verification. We check first-order PMP conditions only. Verifying second-order sufficiency (conjugate point condition or definiteness of the shooting Jacobian) is a possible extension.

5.3. Outlook. The encouraging results open several directions: using the computed PMP solution to seed value-function approximations (e.g., training a neural network near the optimal trajectory), extending the method to controlled diffusions using the companion stochastic framework, and embedding the solver within a hierarchical or model-predictive control scheme. An ablation study quantifying the marginal contribution of each adaptive component (time, PA, δ) would further strengthen the empirical case.

6. Conclusion. We have presented a Pontryagin-based numerical solver for deterministic optimal control problems in Bolza form. The method constructs a concave piecewise-affine bundle surrogate of the Hamiltonian, smooths it via a log-sum-exp formula to obtain a C^∞ Hamiltonian H_δ , and solves the resulting symplectic Euler TPBVP by damped Newton iteration. A unified adaptive outer loop controls three distinct error sources—time discretization, PA modeling, and smoothing bias—via computable a posteriori indicators.

Three benchmark problems (LQR, bang–bang double integrator, and Dubins-type car) illustrate the solver’s workflow: the adaptive mesh concentrates intervals near control switches, the bundle grows only where the PA gap is significant, and the smoothing parameter is annealed as the solution stabilizes.

The present paper is an algorithmic note. The solver is designed for moderate-dimensional problems where the Hamiltonian is accessed via an oracle, the control set is compact, and a reasonable initial guess is available. State constraints are handled by an algorithmic viability filter (tangential Hamiltonian H_K), not by a full state-constrained maximum principle. Extensions to stochastic dynamics, hybrid systems, and rigorous convergence analysis are deferred to future work.

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Appendix A. Reproducibility and Additional Results. All code and data needed to reproduce the numerical experiments are provided in the supplementary material. This includes the solver implementation and scripts for generating the LQR, double integrator, and Dubins car results and plots.

TABLE 2

Optimal cost/time comparison with known reference solutions for Example 1 (LQR) and Example 2 (minimum-time). No analytical reference is available for Example 3.

Example	Reference value	Solver result	Error (abs.)
1: LQR	$J^* = 1.0740$	1.0742	2×10^{-4}
2: Min-time	$T^* = 1.0000$	0.9999	1×10^{-4}
3: Dubins car	(N/A)	0.782	—